Data Handling: Import, Cleaning and Visualisation

Exercise/Workshop 1: Tools, Working with Text Files

Prof. Dr. Ulrich Matter
(University of St. Gallen)

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Prerequisites

- Install the latest R version. R is open source and freely available from here: https://stat.ethz.ch/CRAN/. Click on the respective link for your operating system (Windows, Mac, Linux), and follow the instructions.
- Once R is installed, install RStudio from here: https://www.rstudio.com/products/rstudio/download/#download. Under "Installers for Supported Platforms", click on the respective installer for your operating system (Windows, Mac, etc.) and follow the instructions.

Exercises

Exercise A: Setting up a Working Environment

1. Open RStudio and get familiar with the file browser pane on the lower right. Navigate to a folder on your hard disk in which you want to work throughout this course (and store all the code you write in this course).
2. Use the 'New Folder'-button to create a new folder. Name this new folder `r_course`.
3. You should see the new folder listed in the file browser. Click on it to navigate to its contents (so far empty). Now, click on the 'More' button and select 'Set as Working Directory' in the drop-down menu.
4. Again, use the 'New Folder'-button in order to create two new folders called `data` and `code`.
5. Finally, click on the project button in the top-right corner of the RStudio window (

) and select 'New Project' in the drop-down menu. In the pop-up window, select 'Existing directory', browse to and select your `r_course` folder, then click 'Create Project'.

Now you know how to set up a meaningful basic folder structure and working environment for an R project. The next exercise teaches you how to write R scripts in this environment.

Exercise B: R Scripts

1. Switch to the R console and type the following line of code and hit enter (see example from above).

```r
print("Hello world")
```

You should see the words `"Hello world"` printed on screen. This is the usual way of working with R in an interactive session. However, as pointed out above, in most circumstances it makes sense to write the R code to an R script (in order to store and document it) and then execute the code from there.

2. In the RStudio Menu-bar select `File/New File/R Script` to create a new file, shown/opened in the Script pane.
3. Type `print("Hello world")` to the first line of the script, and click on 'Run' (

) to execute the code in the console.

4. Save the file as `hello_world.R` (`File/Save As...`) in the sub folder `code` (created in the previous exercise).

5. Type the following command into the command line and hit enter:

```
source("code/hello_world.R")
```

```
## [1] "Hello world"
```

Now you know how to execute R code directly in the console (interactive session), how to execute lines of code written to an R script, as well as how to execute the entire R script (stored on disk) from the command line.